

Objective-1: Check ACID compliances in DBMS using Transaction in MySQL.

Environment Creation:

```
CREATE TABLE bank_account (  
    account_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    balance DECIMAL(10,2)  
);
```

Data Insertion:

```
INSERT INTO bank_account VALUES  
(1, 'Md. Golam Hossain', 10000),  
(2, 'Md. Golam Rabbani', 8000);
```

Hands-on:

- open command line	
- go to mysql bin directory	cd C:\xampp\mysql\bin
- run mysql client	mysql -u root
- do the same for another transaction section	

Run-queries:

TXN-1	TXN-2
Atomicity	
BEGIN; UPDATE bank_account SET balance = balance - 5000 WHERE account_id = 1; UPDATE bank_account SET balance = balance + 5000 WHERE account_id = 2;	BEGIN; UPDATE bank_account SET balance = balance - 5000 WHERE account_id = 1; ROLLBACK;

COMMIT;	
Consistency	
SELECT SUM(balance) AS total_balance FROM bank_account;	SELECT SUM(balance) AS total_balance FROM bank_account;
Isolation	
START TRANSACTION; SELECT * FROM lab9.bank_account; UPDATE lab9.bank_account SET balance = balance - 1000 WHERE account_id = 2; SELECT * FROM lab9.bank_account; UPDATE lab9.bank_account SET balance = balance + 1000 WHERE account_id = 1; SELECT * FROM lab9.bank_account; COMMIT;	START TRANSACTION; SELECT * FROM lab9.bank_account; UPDATE lab9.bank_account SET balance = balance - 2000 WHERE account_id = 1; SELECT * FROM lab9.bank_account; UPDATE lab9.bank_account SET balance = balance + 2000 WHERE account_id = 2; SELECT * FROM lab9.bank_account; COMMIT;
Durability	
START TRANSACTION; SELECT * FROM lab9.bank_account; UPDATE lab9.bank_account SET balance = balance - 1000 WHERE account_id = 2; SELECT * FROM lab9.bank_account; ROLLBACK;	Restart Xampp at this point and see.